

Phase -1.4 — Actors, Roles & User Stories

Objective

Before designing APIs or databases, we must understand **who interacts with the system** and **what they want to accomplish**.

This phase converts business requirements into user-centric workflows.

User Stories are intentionally written from the user's perspective rather than the developer's perspective.

The format followed throughout this document is:

As a `<role>`, I want to `<goal>` so that `<business value>`.

System Actors

The system contains three primary actors.

Customer (User)

A registered customer who purchases event tickets.

Permissions include:

- Register
 - Login
 - Browse events
 - View seat availability
 - Book seats
 - Make payments
 - View booking history
 - Logout
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Administrator

An internal user responsible for managing the platform.

Permissions include:

- Create events
- Update events
- Delete events
- Configure seating
- Monitor bookings
- View analytics
- Manage inventory

Administrators inherit every permission available to normal users.

System

The system itself performs automated operations without human interaction.

Examples include:

- Expiring held seats
- Processing delayed jobs
- Writing logs
- Collecting metrics
- Refreshing caches
- Running scheduled background tasks

Although not a human user, the System is treated as an actor because it initiates workflows.

User Stories — Authentication

US-AUTH-01

As a visitor,

I want to create an account,

so that I can purchase tickets.

US-AUTH-02

As a registered user,

I want to log into my account,

so that I can access protected features.

US-AUTH-03

As an authenticated user,

I want to refresh my session,

so that I do not need to log in repeatedly.

US-AUTH-04

As an authenticated user,

I want to log out,

so that my account remains secure.

User Stories — Events

US-EVENT-01

As a visitor,

I want to browse upcoming events,

so that I can decide what interests me.

US-EVENT-02

As a visitor,

I want to view event details,

so that I can learn more before booking.

US-EVENT-03

As a visitor,

I want to see seat availability,

so that I know which seats are still available.

User Stories — Booking

US-BOOK-01

As an authenticated user,

I want to reserve an available seat,

so that no one else can purchase it while I complete payment.

US-BOOK-02

As an authenticated user,

I want to complete payment,

so that my booking becomes permanent.

US-BOOK-03

As an authenticated user,

I want to receive confirmation,

so that I know my booking was successful.

US-BOOK-04

As an authenticated user,

I want to view my previous bookings,

so that I can manage my tickets.

US-BOOK-05

As an authenticated user,

I expect that if payment fails,

my reserved seat becomes available again.

User Stories — Administration

US-ADMIN-01

As an administrator,

I want to create events,

so that tickets become available for purchase.

US-ADMIN-02

As an administrator,

I want to configure seat layouts,

so that customers can select seats.

US-ADMIN-03

As an administrator,

I want to update event information,

so that customers always see accurate details.

US-ADMIN-04

As an administrator,

I want to delete unpublished events,

so that outdated information is removed.

US-ADMIN-05

As an administrator,

I want to monitor bookings,

so that I understand ticket sales and occupancy.

User Stories — Background Processing

US-SYS-01

As the System,

I want to automatically release expired seat holds,

so that inventory becomes available again.

US-SYS-02

As the System,

I want to send booking confirmations asynchronously,

so that user requests complete quickly.

US-SYS-03

As the System,

I want to retry failed background jobs,

so that temporary failures do not permanently lose work.

User Stories — Observability

US-OPS-01

As an operator,

I want structured logs,

so that application failures can be diagnosed efficiently.

US-OPS-02

As an operator,

I want request metrics,

so that I can monitor system performance.

US-OPS-03

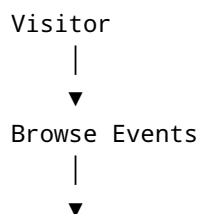
As an operator,

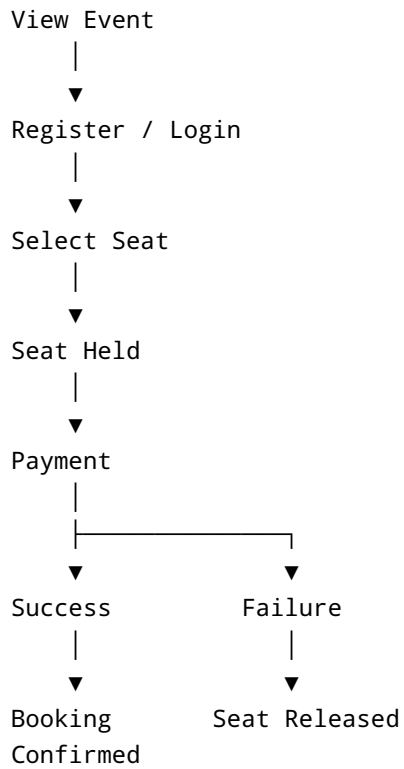
I want health endpoints,

so that deployment platforms can determine application health.

Primary User Journey

The most important workflow in the application is the ticket purchase lifecycle.

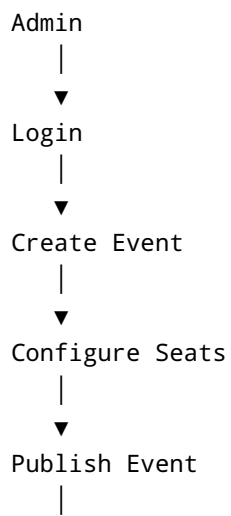




Every feature implemented throughout the project should ultimately support this journey.

Secondary User Journey

Administrator Workflow



Acceptance Criteria

Each user story must have measurable acceptance criteria.

Example:

US-BOOK-01

A booking request is considered successful only if:

- The user is authenticated.
- The seat exists.
- The seat is AVAILABLE.
- Exactly one booking is created.
- The seat transitions to HELD.
- A payment timeout is recorded.
- The operation completes within a single transaction.

Acceptance criteria become the foundation for future integration tests.

Deliverables

At the completion of this phase, the project defines:

- Every actor interacting with the platform.
- Every role and permission.
- Every major workflow.
- The complete ticket purchase lifecycle.
- Administrator workflows.
- System-initiated workflows.
- Acceptance criteria for future implementation.

The next phase, **Phase -1.5 — High-Level System Architecture**, transforms these user interactions into the software architecture that will power the application.